# Numerical Design and Performance Mapping of a Micro Cold-Gas Thruster

Team Repulsor Mark 1
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Course AA374

# The Computational Bottleneck in Micro-Propulsion

## The Constraint

Miniaturisation of spacecraft demands **micro cold-gas thrusters** for precise attitude control.

Strict physical limits dictate throat areas on the order of just **1 $mm^2$**.
Requires mapping of **transonic acceleration, shock wave formation,** and **deep-space vacuum expansion**.

## The Bottleneck

Commercial 2D/3D CFD software is **computationally prohibitive** for rapid parametric design iterations.

Standard analytical **isentropic equations** fundamentally **fail to capture shock discontinuities** and entropy-generating phenomena.

## The Solution

A custom **quasi-1D finite-volume solver** built from scratch.
Bridges the gap: **computationally efficient** yet **highly accurate** for **shock-capturing** via **advanced Total Variation Diminishing (TVD) flux limiters**.

# Mathematical Architecture and Discretization

## Physics Engine

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} = S$$

$$\begin{bmatrix} \rho A \\ \rho u A \\ E A \end{bmatrix} \quad \begin{bmatrix} \rho u A \\ \rho u^2 A + pA \\ u(EA + pA) \end{bmatrix} \quad \begin{bmatrix} 0 \\ \frac{p \partial A}{\partial x} \\ 0 \end{bmatrix} \begin{array}{l} \leftarrow \text{Mass } ([\rho A]) \\ \leftarrow \text{Momentum } ([\rho u A]) \\ \leftarrow \text{Energy } ([EA]) \end{array}$$

### Lax-Friedrichs

The baseline method. Simple, robust, but inherently highly diffusive.

### HLLC (Approximate Riemann)

Resolves the three-wave structure, including contact discontinuities.

### MUSCL-HLLC

Second-order spatial accuracy utilising a MinMod flux limiter to prevent non-physical oscillations.

# The Numerical Arsenal: Evolution of Shock Capturing

## LAX-FRIEDRICHS

*The Baseline*

A 1st-order finite volume scheme. It enforces stability by introducing artificial viscosity scaled to the maximum local wave speed.

Highly robust and computationally forgiving, but excessively diffusive. It smears sharp shockwaves into wide, inaccurate gradients, failing to resolve the true physics of the discontinuity.

## HLLC

*Approximate Riemann Solver*

An advanced 1st-order solver. It resolves the exact 3-wave structure of the flow field, explicitly tracking the contact discontinuity to better estimate wave speeds.

Significantly reduces numerical diffusion, revealing actual shock geometry. However, this precision comes at a cost, occasionally introducing minor, non-physical oscillations near the shock front.

## MUSCL-HLLC

*The Advanced Standard*

Applies 2nd-order spatial reconstruction to the HLLC framework. It utilises limited gradients paired with a strictly mathematical MinMod TVD Limiter to filter the Gibbs phenomenon.

The definitive synthesis. It cures the volatility of the Riemann solver, capturing shockwaves as razor-sharp cliffs while maintaining absolute numerical stability and high-fidelity representation of smooth expansion regions.

# Defining the Computational Boundaries

## Nozzle Geometries

### Double-Cone (Piecewise Linear)

Features a distinct discontinuity in the area gradient at the throat.

### Anderson Parabolic (Smooth)

Defined by a $C^1$-continuous quadratic profile:

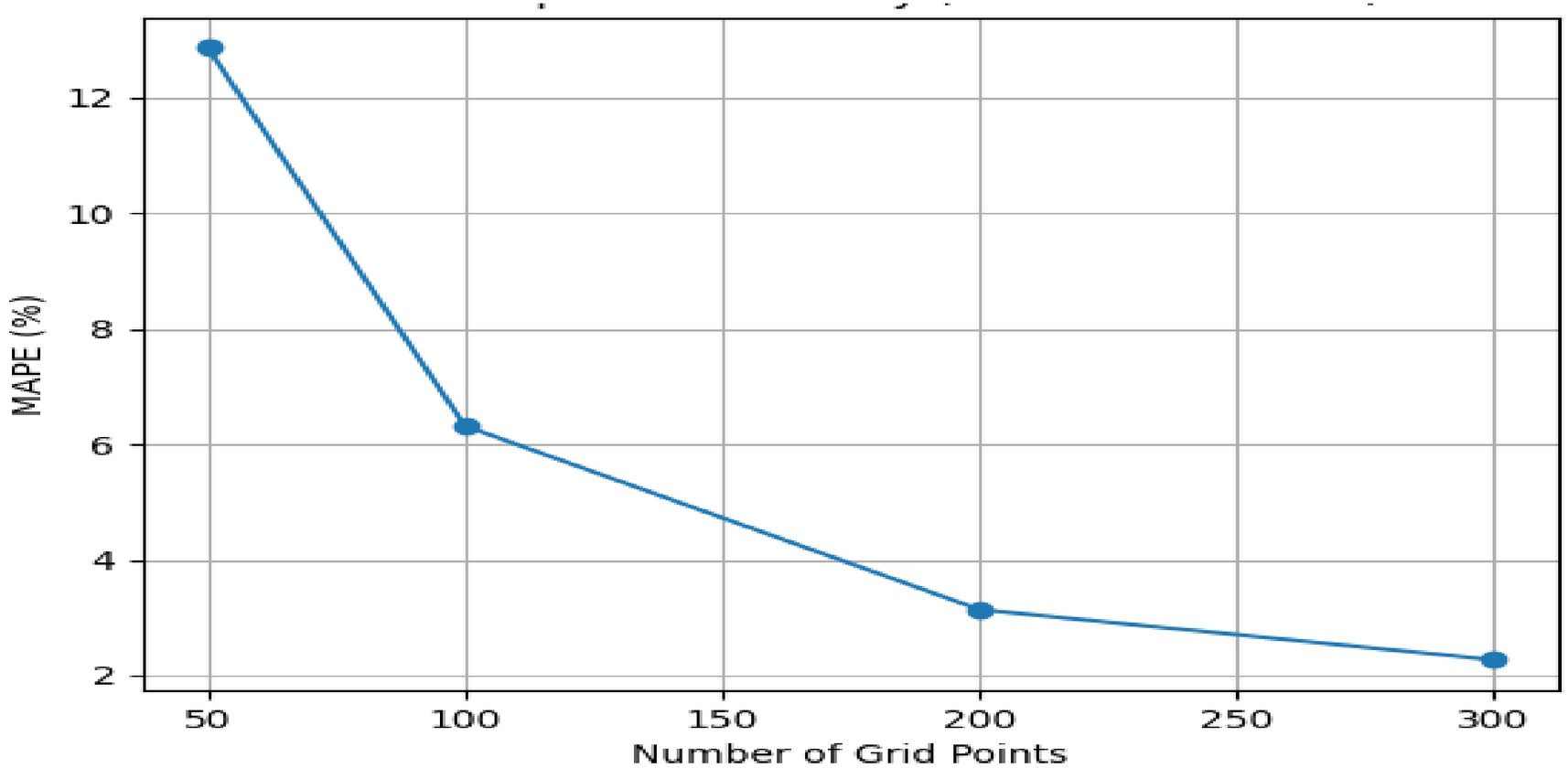
$$\frac{A(x)}{A^*} = 1 + 2.2\left(\frac{x}{L} - 1.5\right)^2$$

## Thermodynamic Conditions

- **Stagnation Inlet: 3 bar ($3 \times 10^5$ Pa) at 600 K,** mimicking a heated micro cold-gas chamber.
- **Outlet:** Characteristic **non-reflecting boundaries** for subsonic flow; extrapolation for supersonic flow.
- **Variable back pressure** ($p_{back}$) to evaluate over-expanded to perfectly-expanded deep space vacuum regimes.

# Verification Phase I: Spatial Convergence

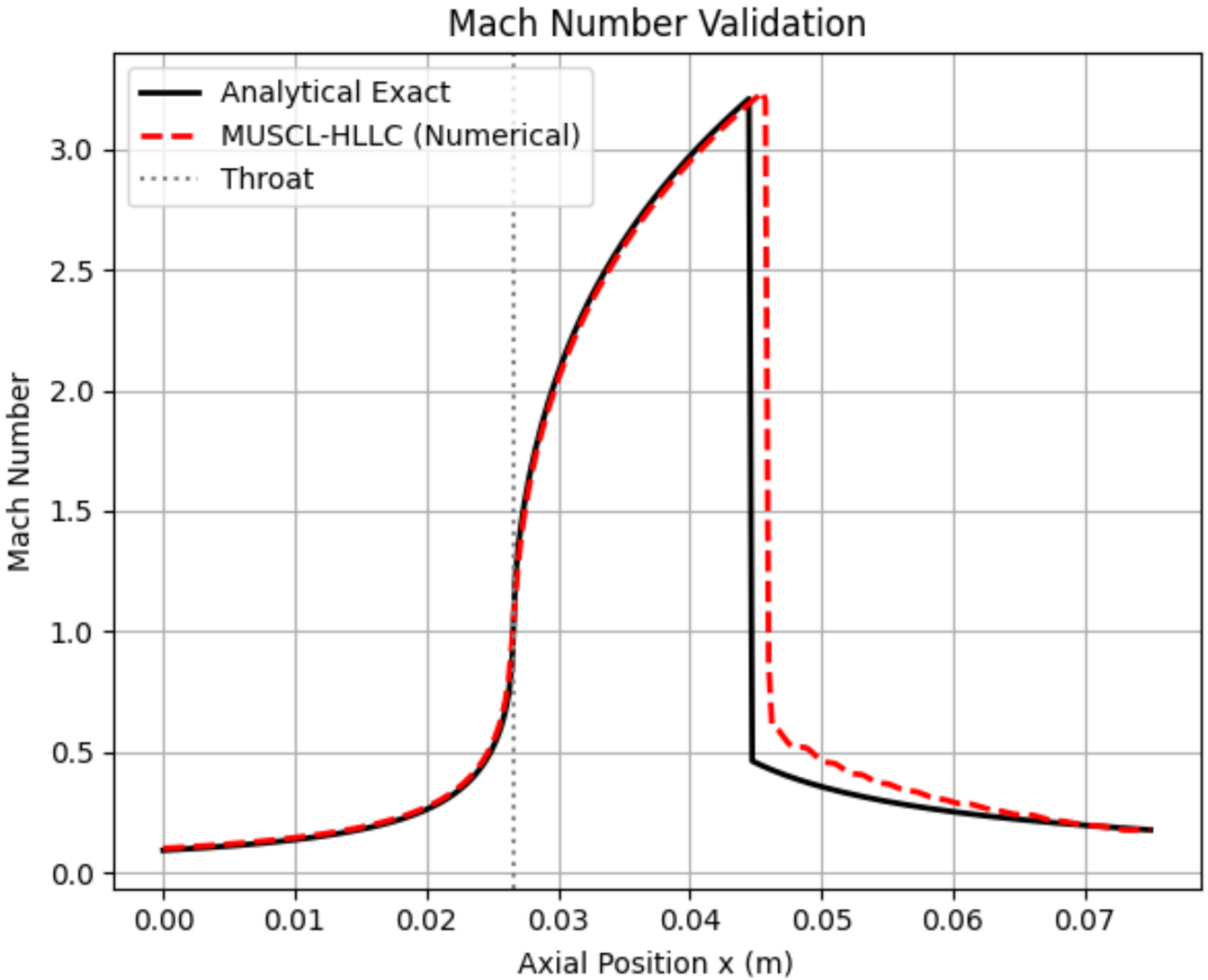
Optimal grid size was established to perfectly balance computational efficiency with high-fidelity solution accuracy.




Data Insight:

Asymptotic reduction in Mean Absolute Percentage Error (MAPE) as grid resolution increases proves the solution is mathematically sound and grid-independent.

# Verification Phase II: Exact Analytical Validation




## MUSCL-HLLC Performance

The solver perfectly tracks the isentropic line and captures the normal shock discontinuity without breaking.

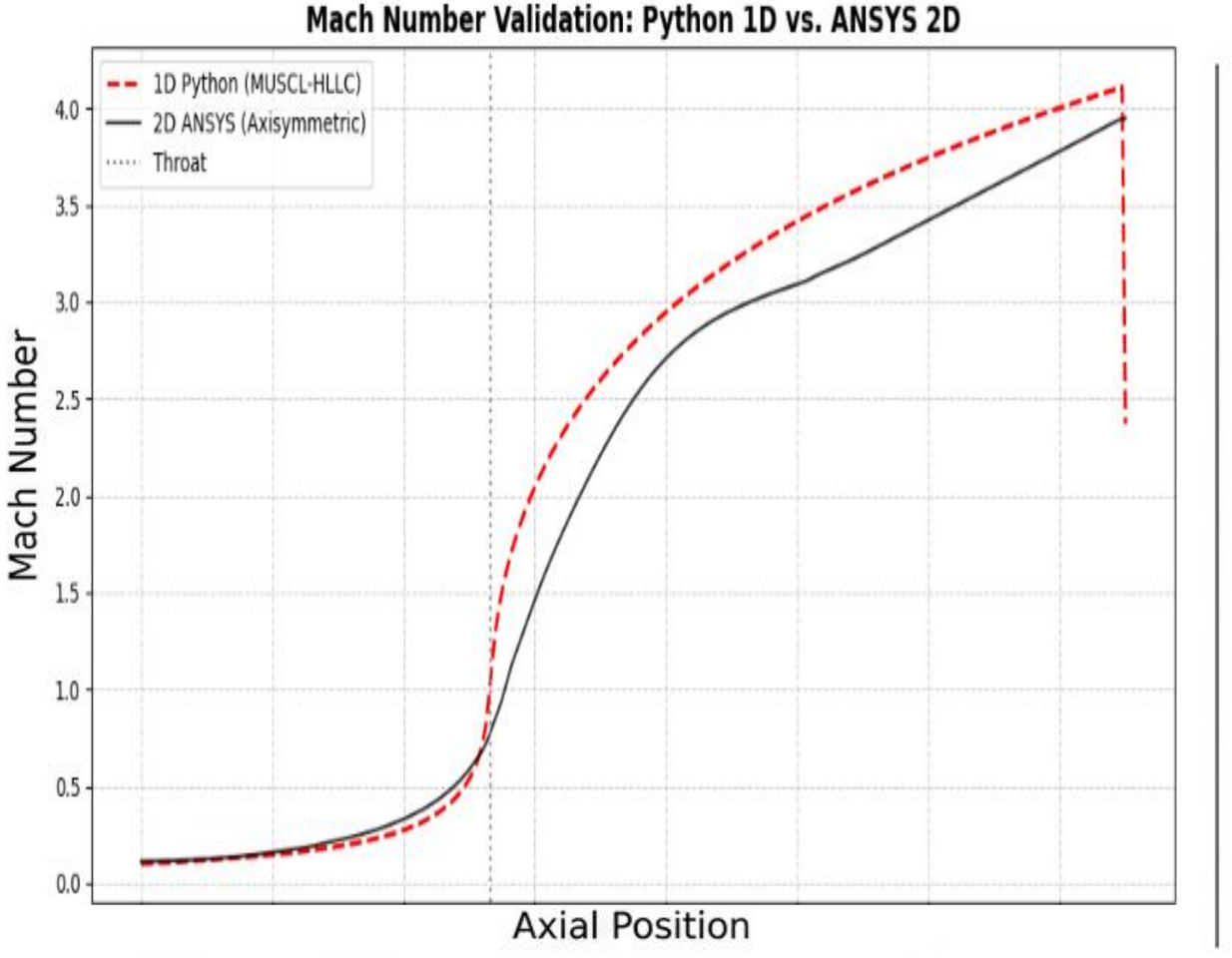
Mean Absolute Percentage Error (MAPE):

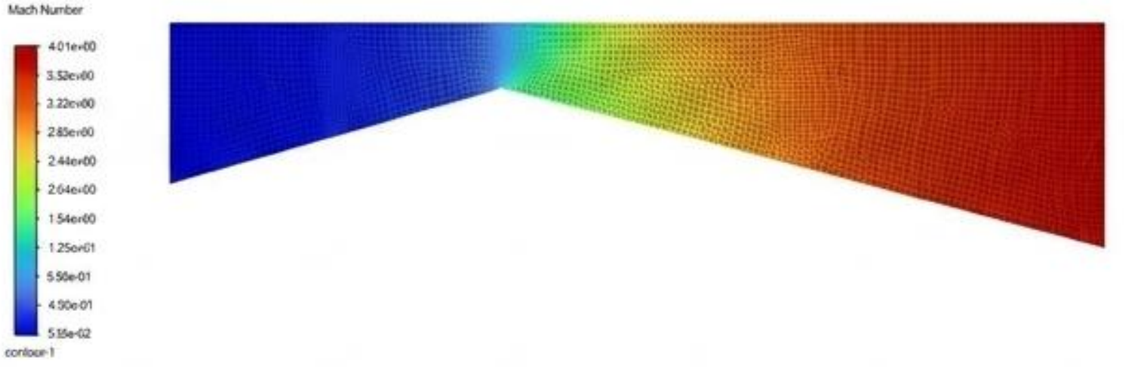
**2.24%**

Maximum error remains strictly localized precisely at the shock due to its discontinuous nature.

# Verification Phase III: Commercial 2D CFD Comparison

The custom solver successfully captures the dominant flow physics—transonic acceleration and supersonic expansion—at a fraction of the computational cost.



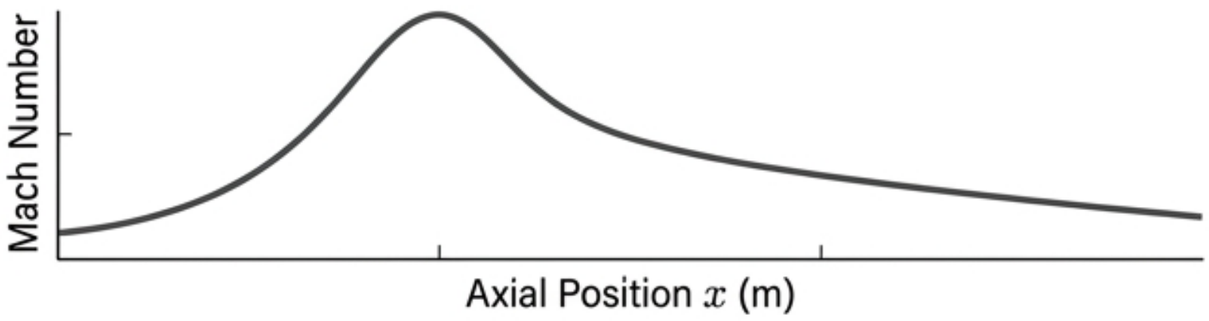




Note: The 12.60% MAPE deviation is an expected limitation of 1D modelling, arising from 2D boundary layer growth and radial gradients not present in quasi-1D space.

# Stratification of Flow Regimes

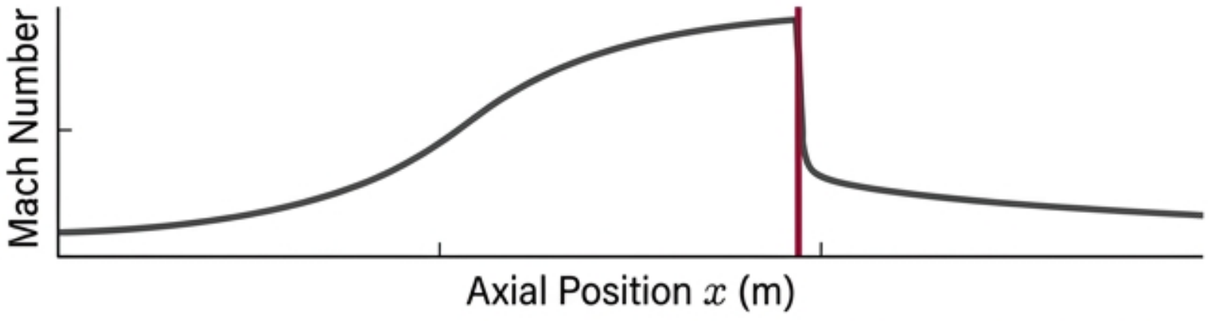
## Fully Subsonic

Acceleration towards the throat, smooth deceleration in the diverging section.



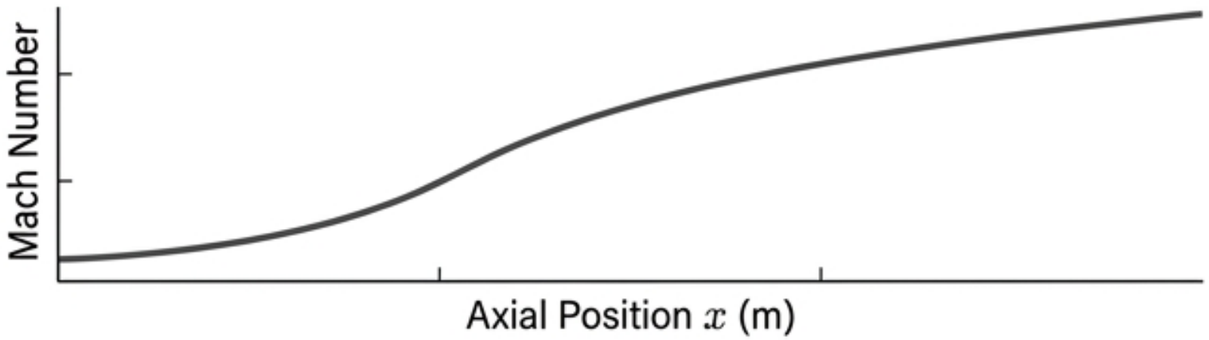

## Shock-Containing

Supersonic acceleration post-throat abruptly terminated by a normal shock discontinuity.




## Fully Supersonic

Monotonic acceleration simulating perfectly expanded deep space vacuum conditions.



# Diagnostic Consensus: The Scheme Showdown

| | Shock Resolution | Spatial Accuracy | Stability |
|---|---|---|---|
| Lax-Friedrichs (Baseline) | Highly smeared/diffused | 1st-Order | Robust, no limiters required |
| HLLC (Riemann Solver) | Sharp, but susceptible to minor oscillations | 1st-Order | Wave speed dependent |
| MUSCL-HLLC (Optimal) | Exceedingly sharp, minimal diffusion | 2nd-Order reconstruction | Highly stable via MinMod TVD limiting |

# Scaling to Physical Reality: Thruster Performance

## Quadratic Nozzle

| Propellant | Absolute Thrust (mN) | Specific Impulse (s) |
|---|---|---|
| Nitrogen | 447.51 | 94.40 |
| Helium | 451.14 | 234.76 |

## Double-Cone Nozzle

| Propellant | Absolute Thrust (mN) | Specific Impulse (s) |
|---|---|---|
| Nitrogen | 495.57 | 100.71 |
| Helium | 481.90 | 244.06 |

## The Core Engineering Trade-Off

- Helium achieves superior Specific Impulse (**~244 s**) due to lower molecular weight and higher wave speed.
- Nitrogen generates **higher absolute thrust** (**~495 mN**) driven by higher density and mass flow rate.

# Conclusion and Engineering Value

## Optimal Fidelity

The 2nd-order MUSCL-HLLC scheme provides the ultimate balance of high-resolution shock capturing and numerical stability.

## Performance Mapping

The numerical framework successfully mapped critical thruster parameters across multiple geometric and thermodynamic boundaries.

## Design Velocity

The custom 1D solver acts as a highly reliable, mathematically proven testbed for preliminary space propulsion design—operating at a fraction of the computational cost of multi-dimensional CFD.

# THANK YOU !